

4.1

New circuit
analysis
techniques

A direct application of Kirchhoff's current and voltage laws using the principles discussed in Chapter 3 can solve many circuit problems. However, there are a variety of techniques, all based on these two laws, that can simplify circuit analysis. The main techniques, to be introduced in this chapter, are:

- Mesh analysis.
- Nodal analysis.
- Thévenin's theorem.
- Norton's theorem.

Each of these techniques has particular strengths aimed at solving particular types of circuit problem. In this way, what would be laborious using one method can be straightforward using another. Familiarization with all the different methods will enable you to choose the method which best suits a particular problem. This simplifies circuit solution and makes less work overall!

4.2

Kirchhoff's laws
and network
solution

Kirchhoff's laws can be applied to network solution in any of the following ways:

1. By direct application to the network in conjunction with Ohm's law.
2. By indirect application to the network in conjunction with the manipulation of the component resistances.
3. By direct application to the network resulting in solution by simultaneous equations.

These statements appear to be most complicated, but the following series of examples will illustrate the forms of application of the laws to network solution. The form that ought to be most obvious is the first form, in which the laws are directly applied; curiously this form of solution tends to be so obvious that it is all too often neglected, as will be illustrated.

Example 4.1

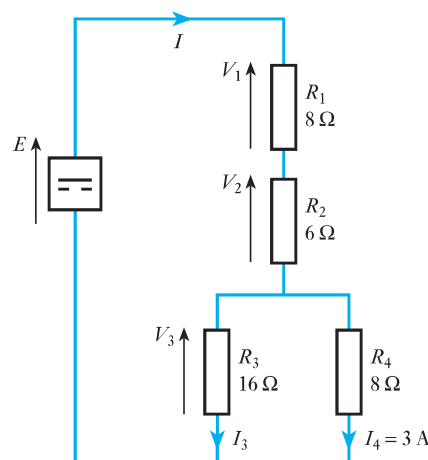
For the network shown in Fig. 4.1, determine the supply current and the source e.m.f.

Since R_3 and R_4 are in parallel

$$V_3 = I_4 R_4 = 3 \times 8 = 24 \text{ V} = I_3 R_3 = I_3 \times 16$$

$$I_3 = \frac{24}{16} = 1.5 \text{ A}$$

Fig. 4.1 Circuit diagram for Example 4.1



By Kirchhoff's first law

$$I = I_3 + I_4 = 1.5 + 3 = 4.5 \text{ A}$$

Also $V_1 = IR_1 = 4.5 \times 8 = 36 \text{ V}$

$$V_2 = IR_2 = 4.5 \times 6 = 27 \text{ V}$$

By Kirchhoff's second law

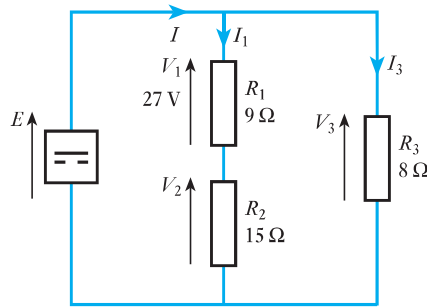
$$E = V_1 + V_2 + V_3 = 36 + 27 + 24 = 87 \text{ V}$$

This is not the only form of solution to the given problem. For instance, the supply current could have been derived directly from I_3 by applying the current-sharing rule, or the source e.m.f. could have been derived from the product of the supply current and the total effective resistance which could have been determined – but the direct solution is readily available without the need to resort to such devices. The following two examples illustrate again the availability of a direct approach to network problems.

Example 4.2

Given the network shown in Fig. 4.2, determine I_1 , E , I_3 and I .

Fig. 4.2 Circuit diagram for Example 4.2



$$I_1 = \frac{V_1}{R_1} = \frac{27}{9} = 3 \text{ A}$$

$$V_2 = I_1 R_2 = 3 \times 15 = 45 \text{ V}$$

$$E = V = V_1 + V_2 = 27 + 45 = 72 \text{ V}$$

$$I_3 = \frac{V}{R_3} = \frac{72}{8} = 9 \text{ A}$$

$$I = I_1 + I_3 = 3 + 9 = 12 \text{ A}$$

Example 4.3

For the network shown in Fig. 4.3, the power dissipated in R_3 is 20 W. Calculate the current I_3 and hence evaluate R_1 , R_3 , I_1 , I_2 and V .

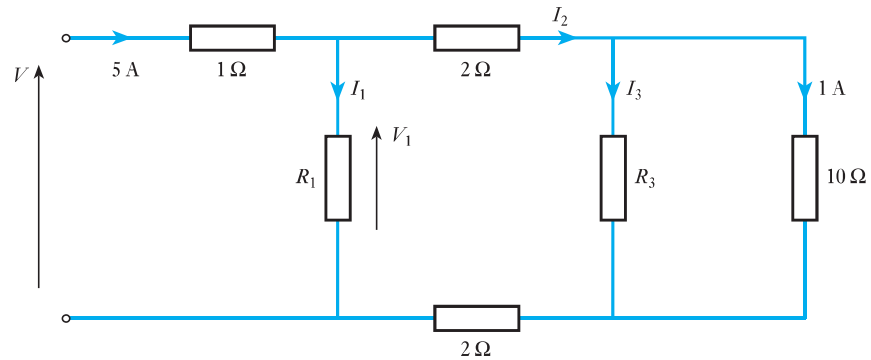
Potential difference across the 10Ω resistor is $1 \times 10 = 10 \text{ V}$. For resistor R_3 ,

$$P = 20 \text{ W} = 10 \times I_3$$

Hence $I_3 = \frac{20}{10} = 2 \text{ A}$

$$P = I_3^2 R_3 = 20$$

Fig. 4.3 Circuit diagram for Example 4.3



$$\text{hence } 20 = 2^2 \times R_3$$

$$R_3 = 5 \, \Omega$$

$$I_2 = 2 + 1 = 3 \, \text{A}$$

Potential difference across each of the two $2 \, \Omega$ resistors is $3 \times 2 = 6 \, \text{V}$. Thus

$$V_1 = 6 + 10 + 6 = 22 \, \text{V}$$

$$I_1 = 5 - 3 = 2 \, \text{A}$$

$$R_1 = \frac{V_1}{I_1} = \frac{22}{2} = 11 \, \Omega$$

Potential difference across the $1 \, \Omega$ resistor is $5 \times 1 = 5 \, \text{V}$, hence

$$V = 5 + 22 = 27 \, \text{V}$$

This last example in particular illustrates that a quite complicated network can readily be analysed by this direct approach. However, it is not always possible to proceed in this way, either because most of the information given relates to the resistances or because there is insufficient information concerning any one component of the network.

An instance of the information being presented mainly in terms of resistance is given in Example 4.4 and it also brings us to the second form of application of Kirchhoff's laws.

Example 4.4

For the network shown in Fig. 4.4, determine the supply current and current I_4 .

In essence this network consists of three parts in series, but one of them comprises R_3 and R_4 in parallel. These can be replaced by an equivalent resistance, thus

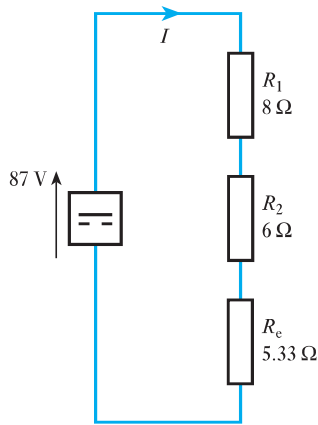
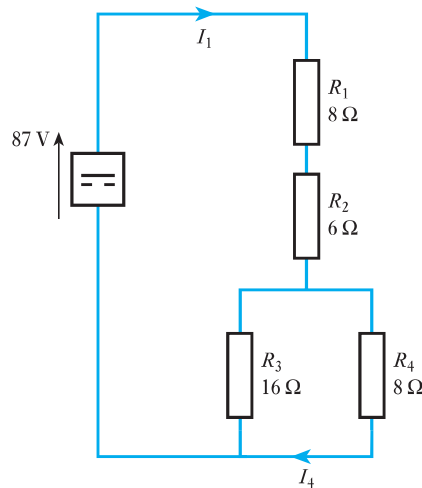
$$R_c = \frac{R_3 R_4}{R_3 + R_4} = \frac{16 \times 8}{16 + 8} = 5.33 \, \Omega$$

Replacing R_3 and R_4 by R_c , the network becomes that shown in Fig. 4.5.

Now that the network has been reduced to a simple series circuit the total effective resistance is

$$R = R_1 + R_2 + R_c = 8 + 6 + 5.33 = 19.33 \, \Omega$$

$$I = \frac{V}{R} = \frac{87}{19.33} = 4.5 \, \text{A}$$

Fig. 4.4 Circuit diagram for Example 4.4**Fig. 4.5** Circuit diagram for Example 4.4

Reverting now to the original network,

$$I_4 = \frac{R_3}{R_3 + R_4} \cdot I = \frac{16}{16 + 8} \times 4.5 = 3 \text{ A}$$

This example compares with Example 4.1 and the figures are in fact the same. However, in this second instance the given voltage and current information stemmed from the source and not from the load, hence the emphasis of the calculation lay in dealing with the resistances of the network. The calculation was based on network reduction, i.e. by replacing two or more resistors by one equivalent resistor. A further example of this approach is given below, in which two instances of network reduction transform the problem into a form that can be readily analysed.

Example 4.5

Determine V_{AB} in the network shown in Fig. 4.6.

This is quite a complex network. However, there are two instances of parallel resistors that may be replaced by equivalent resistors. For the 10 Ω and 15 Ω resistors

$$R = \frac{10 \times 15}{10 + 15} = 6 \Omega$$

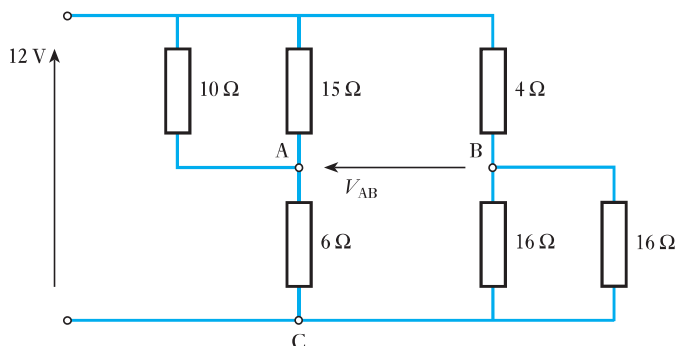
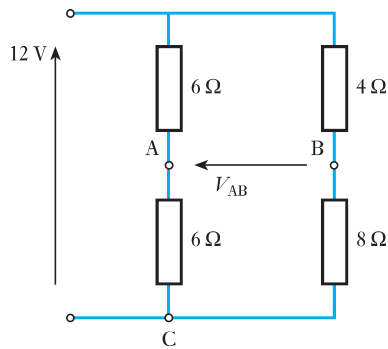
Fig. 4.6 Circuit diagram for Example 4.5

Fig. 4.7 Circuit diagram for Example 4.5



For the two $16\ \Omega$ resistors in parallel

$$R = \frac{16 \times 16}{16 + 16} = 8\ \Omega$$

If these equivalent values are inserted into the network, the network transforms into that shown in Fig. 4.7. Thus

$$V_{AC} = \frac{6}{6 + 6} \times 12 = 6\ \text{V}$$

and
$$V_{BC} = \frac{8}{4 + 8} \times 12 = 8\ \text{V}$$

$$V_{AB} = V_{AC} - V_{BC} = 6 - 8 = -2\ \text{V}$$

Having now observed the two methods of analysis being demonstrated, you may well wonder how to tell when each should be used. As a general rule, if the information given concerns the voltage or the current associated with one or more components of the network, then you would apply the first form of approach. However, if the information given concerns the supply voltage or current, then you would try to apply the second form of approach by network reduction. This is not always possible because resistors may be connected in a manner that is neither series nor parallel – such an arrangement is shown in Fig. 4.8.

Example 4.6

For the network shown in Fig. 4.8, calculate the currents in each of the resistors.

Fig. 4.8 Circuit diagram for Example 4.6

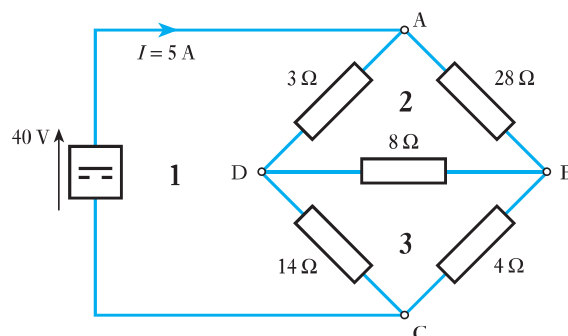
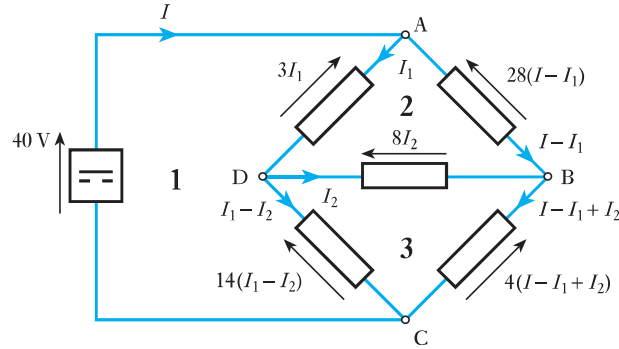


Fig. 4.9 Circuit diagram for Example 4.6



In this network the resistors are neither in series nor in parallel and therefore a more difficult method of analysis must be employed. Let the current in the $3\ \Omega$ resistor be I_1 and therefore by Kirchhoff's first law, the current in the $28\ \Omega$ resistor is $I - I_1$. Further, let the current in the $8\ \Omega$ resistor flowing from D to B be I_2 . It follows that the current in the $14\ \Omega$ resistor is $I_1 - I_2$ while that in the $4\ \Omega$ resistor is $I - I_1 + I_2$. The resulting volt drops are shown in Fig. 4.9.

Applying Kirchhoff's second law to loop 1 (comprising source to ADC):

$$40 = 3I_1 + 14(I_1 - I_2)$$

$$40 = 17I_1 - 14I_2 \quad (a)$$

Applying Kirchhoff's second law to loop 2 (ABD):

$$0 = 28(I - I_1) - 8I_2 - 3I_1$$

$$= 28I - 31I_1 - 8I_2$$

But $I = 5\text{ A}$

Therefore

$$140 = 31I_1 + 8I_2 \quad (b)$$

$$(a) \times 4 \quad 160 = 68I_1 - 56I_2 \quad (c)$$

$$(b) \times 7 \quad 980 = 217I_1 + 56I_2 \quad (d)$$

$$(c) + (d) \quad 1140 = 285I_1$$

$$I_1 = 4\text{ A in } 3\ \Omega \text{ resistor}$$

Substituting in (b),

$$140 = 124 + 8I_2$$

$$I_2 = 2\text{ A in } 8\ \Omega \text{ resistor}$$

Hence current in $28\ \Omega$ resistor is

$$5 - 4 = 1\text{ A}$$

current in $14\ \Omega$ resistor is

$$4 - 2 = 2\text{ A}$$

and current in $4\ \Omega$ resistor is

$$5 - 4 + 2 = 3\text{ A}$$

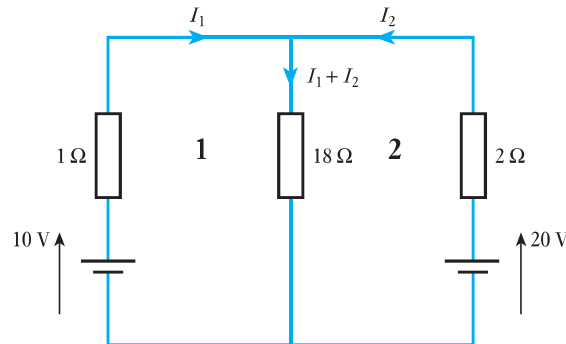
This form of solution requires that you proceed with great caution, otherwise it is a simple matter to make mistakes during the mathematical processes. However, in the instance given, it is necessary to involve such an analysis; had a different current been given in this example, such a solution would not have been required since it would then have been possible to achieve a solution by applying the first approach, i.e. directly applying Kirchhoff's laws.

If two parallel e.m.f.s appear in a network as exemplified by Fig. 4.10, it might again be necessary to employ the approach using simultaneous equations resulting from the application of Kirchhoff's laws.

Example 4.7

Calculate the currents in the network shown in Fig. 4.10.

Fig. 4.10 Circuit diagram for Example 4.7



Applying Kirchhoff's second law to loop 1:

$$10 = 1I_1 + 18(I_1 + I_2)$$

$$10 = 19I_1 + 18I_2 \quad (a)$$

Applying Kirchhoff's second law to loop 2:

$$20 = 2I_2 + 18(I_1 + I_2)$$

$$20 = 18I_1 + 20I_2 \quad (b)$$

$$(a) \times 10 \quad 100 = 190I_1 + 180I_2 \quad (c)$$

$$(b) \times 9 \quad 180 = 162I_1 + 180I_2 \quad (d)$$

$$(d) - (c) \quad 80 = -28I_1$$

$$I_1 = -2.85 \text{ A}$$

Substituting in (a)

$$10 = -54.34 + 18I_2$$

$$I_2 = 3.57 \text{ A}$$

Current in 18Ω resistor is

$$3.57 - 2.85 = 0.72 \text{ A}$$

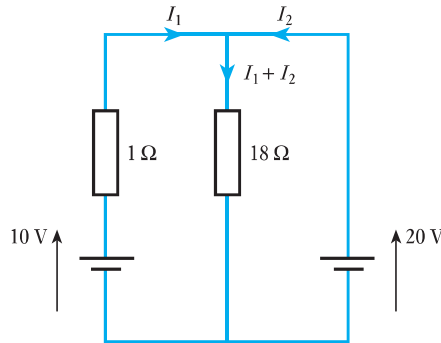
This form of solution is fraught with the danger of mathematical mistakes and therefore should only be employed when all else fails. This section

commenced by stating that the obvious solution is all too easily ignored. Thus if the $2\ \Omega$ resistor were removed from the network shown in Fig. 4.10, it might be overlooked that the 20 V battery is now directly applied to the $18\ \Omega$ resistor and so, knowing the voltage drop across one of the components, it is possible to revert to the first form of analysis as shown in Example 4.8.

Example 4.8

Calculate the currents in the network shown in Fig. 4.11.

Fig. 4.11 Circuit diagram for Example 4.8



Current in $18\ \Omega$ resistor is

$$\frac{20}{18} = 1.1\text{ A}$$

Applying Kirchhoff's second law to the outside loop:

$$20 - 10 = -I_1 \times 1$$

$$I_1 = -10\text{ A}$$

$$I_2 = -(-10) + 1.1 = 11.1\text{ A}$$

Example 4.9

Three similar primary cells are connected in series to form a closed circuit as shown in Fig. 4.12. Each cell has an e.m.f. of 1.5 V and an internal resistance of $30\ \Omega$. Calculate the current and show that points A, B and C are at the same potential.

In Fig. 4.12, E and R represent the e.m.f. and internal resistance respectively of each cell.

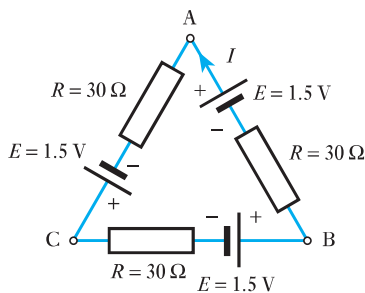


Fig. 4.12 Circuit diagram for Example 4.9

$$\text{Total e.m.f.} = 1.5 \times 3 = 4.5\text{ V}$$

$$\text{Total resistance} = 30 \times 3 = 90\ \Omega$$

$$\therefore \text{Current} = \frac{4.5}{90} = 0.05\text{ A}$$

The volt drop due to the internal resistance of each cell is 0.05×30 , namely 1.5 V . Hence the e.m.f. of each cell is absorbed in sending the current through the internal resistance of that cell, so that there is no difference of potential between the two terminals of the cell. Consequently the three junctions A, B and C are at the same potential.